

Part 4: Comparing TCP and UDP.

Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	• TCP is connection-oriented. • It establishes a connection before normal data transfer. • It provides reliable delivery, unlike UDP, which does not guarantee delivery.
Data Integrity and Ordering	TCP	• TCP delivers the byte stream to the application in order. • It uses sequence numbers to track data. • TCP uses a checksum to detect corruption in TCP segments.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	• Applications where reliable and ordered delivery is important. • Web traffic using HTTP/1.1 and HTTP/2. • Email transfer and access.	• Online gaming where low latency is prioritized. • Live audio/video applications. • Real-time voice communication.
Performance	• Higher communication overhead. • Higher initial latency. • Packet loss can increase delay. • Ordered delivery can add latency. • Better suited when reliable delivery is more important than minimizing latency.	• Lower communication overhead. • Lower initial latency. • No retransmission delays at the UDP layer. • No ordering delays at the UDP layer. • Well suited to latency-sensitive applications